Neural-Symbolic Post-Tonal Composition Assistance with Pitch-Class Set, Serial, and Rhythmic-Profile Constraints

融合音级集合、序列与节奏轮廓约束的神经符号后调性作曲辅助方法

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Abstract

Explicit control over post-tonal materials remains underrepresented in symbolic music generation, while evaluation on copyrighted contemporary repertoire is difficult to reproduce legally. We present a score-level neural-symbolic system that conditions a causal Transformer on pitch-class sets, interval vectors, twelve-tone rows and transformations, rhythmic profiles, gestures, instrumentation, voice count, and requested span. A legal rule-generated corpus provides 20,000/2,000/2,000 training, validation, and test fragments. At inference, non-differentiable post-tonal diagnostics rerank four sampled candidates before MusicXML and an analysis report are exported. A controlled comparison on 2,000 aligned test conditions holds the checkpoint, conditions, and first sampled candidate fixed. Relative to single-candidate decoding, four-candidate reranking increased cyclic row-order accuracy from 0.1580 to 0.2350 (favorable mean difference 0.0769; paired bootstrap 95% CI [0.0705, 0.0833]), reduced rhythmic-profile distance from 0.8712 to 0.8159 (0.0553; [0.0493, 0.0618]), and increased heuristic gesture consistency from 0.5246 to 0.5535 (0.0289; [0.0249, 0.0330]). These gains were not uniform: serial aggregate completion decreased from 0.9963 to 0.9892, and overall pitch-class-set coverage showed no clear directional change. All 20 inspected outputs parsed structurally as MusicXML. Because reranking and evaluation share the same diagnostics, the findings establish controllability on the synthetic testbed rather than artistic quality or composer preference. The system is therefore positioned as a reproducible proof of concept for explainable post-tonal score sketching.

Keywords: computer-assisted composition; post-tonal music; symbolic music generation; pitch-class set theory; twelve-tone serialism; constraint-guided decoding; MusicXML

1 Introduction

Post-tonal composition frequently begins from explicit symbolic materials: a pitch-class collection, an interval-class profile, a twelve-tone row and its transformations, a registral trajectory, or a characteristic rhythmic surface. Computer-assisted composition environments have long supported the manipulation of such structures, but recent neural music systems are more often evaluated on tonal continuation, popular-song structure, performance MIDI, or audio generation [2, 3]. Those settings do not directly answer a practical question faced by composers of contemporary score-based music: can a generative model produce an editable sketch while preserving named post-tonal constraints and explaining where it satisfies or violates them?

This question requires a different interface and a different evaluation vocabulary. Pitch-class set theory supplies normal order, prime form, and interval vectors [6, 13]; serial practice requires explicit $P/R/I/RI$ transformations and aggregate tracking; rhythmic and gestural intentions are expressed over score time rather than audio frames. The output must also be inspectable as notation. A fluent token sequence is therefore insufficient if it cannot be converted into a score or if its relation to the requested material remains opaque.

Data legality presents a second challenge. A large fraction of relevant post-1945 repertoire remains copyrighted, and a scraped corpus would be difficult to release or reproduce. We instead use a deterministic procedural corpus as a controlled testbed. The corpus is not intended to imitate a composer or establish historical style. Its purpose is to expose known condition–target relations so that constraint adherence, failure modes, and decoding trade-offs can be measured without distributing protected scores.

We study a condition-prefix Transformer that generates quantized event sequences and a neural-symbolic decoder that reranks sampled candidates using non-differentiable post-tonal diagnostics. The approach is deliberately modest at the neural level: it does not introduce a new attention architecture. Its contribution lies in connecting causal symbolic generation to pitch-class-set, serial, rhythmic, gestural, register, and notation-level checks, and in returning both a MusicXML sketch and a machine-readable explanation.

The study addresses three research questions:

- **RQ1:** With the neural checkpoint and test conditions fixed, does multi-candidate symbolic reranking improve explicit constraint metrics relative to single-candidate sampling?
- **RQ2:** What descriptive patterns appear across the archived condition-removal and fixed-default configurations?
- **RQ3:** Can the resulting fragments be exported reliably as structurally valid MusicXML with inspectable analysis reports?

The main contributions are:

1. a legal and reproducible 24,000-fragment synthetic corpus protocol for score-level post-tonal conditions;
2. a compact condition-prefix/event representation and locally trainable Transformer conditioned on requested spans of 4–16 measures and 2–8 voices;

3. an inference-time candidate-reranking objective grounded in pc-set, serial, rhythmic, gestural, and range diagnostics; and
4. an evaluation and export layer that connects aggregate metrics to MusicXML examples and per-fragment JSON reports.

The resulting claims are intentionally bounded. Quantitative findings concern a synthetic constraint testbed, not stylistic authenticity. Expert judgments of compositional usefulness are planned with the supplied blind-rating package but are not inferred from automatic metrics.

2 Related Work

2.1 Computer-assisted composition and post-tonal representation

Computer-assisted composition has a longer history than data-driven music generation. OpenMusic represents musical objects and transformations in a visual programming environment, allowing a composer to formalize and revise compositional procedures without reducing the task to audio synthesis [2]. The music21 toolkit provides a complementary programmatic layer for symbolic score analysis, transformation, and interchange [4]. These systems establish symbolic inspectability as a practical requirement for compositional software.

Constraint programming offers a second precedent for explicit musical control. Anders and Miranda survey systems that express theories of harmony, rhythm, form, and instrumentation as declarative constraints [1]. This tradition treats a musical rule as an inspectable compositional object rather than as an implicit property of a corpus. Our decoder follows the same preference for explicit diagnostics, while using sampled neural candidates instead of solving a complete constraint-satisfaction problem.

Pitch-class set theory supplies equivalence under transposition and inversion, normal and prime forms, and interval-class vectors for non-tonal pitch organization [6, 13]. Post-tonal theory also treats ordered twelve-tone rows, $P/R/I/RI$ transformations, and aggregates as distinct from unordered set classes. The present system keeps that distinction in both its condition representation and its analysis report.

2.2 Neural symbolic music generation

Neural music generation spans different objectives, representations, and forms of control [3]. The Transformer architecture provides a standard causal sequence model [14], and Music Transformer adapted self-attention to long symbolic performance sequences [10]. Later event representations made meter and beat structure more explicit for expressive pop-piano generation [11]. These studies show that event sequences can support long musical outputs, but their principal corpora and evaluation tasks do not directly measure pitch-class-set adherence, serial transformations, or post-tonal gesture control.

Other symbolic models emphasize polyphonic coordination or latent structure. MuseGAN generates several instrumental tracks in a piano-roll representation and evaluates a rock-music accompaniment setting [5]. MusicVAE uses a hierarchical recurrent decoder to model longer musical sequences in a continuous latent space [12]. These objectives are relevant to multi-part organization and variation, but neither formulation exposes the pitch-class-set and row-form conditions required by the present task.

Our model therefore uses the Transformer as an established sequence generator rather than claiming an architectural advance. Its event vocabulary is intentionally smaller than performance-oriented encodings because the output target is a quantized, editable score fragment. Conditions for pitch, serial order, rhythm, gesture, instrumentation, and span are written into the same symbolic sequence as a prefix.

2.3 Constraint-aware generation

Several systems make neural generation steerable. DeepBach permits positional constraints on notes, rhythms, and cadences during pseudo-Gibbs sampling [8]. Anticipation-RNN incorporates future unary constraints into sequence generation through a backward constraint network and a forward token network [7]. Outside purely neural sampling, MorpheuS uses optimization to preserve recurrent patterns and follow a target tension profile [9]. These methods motivate explicit control, but they address different musical domains and constraint types.

FIGARO is a closer precedent for interpretable conditional generation. It reconstructs symbolic music from learned and expert descriptions, including time-varying high-level features [15]. Our study narrows the control vocabulary to post-tonal compositional materials and adds deterministic analysis after sampling. The resulting comparison asks whether reranking candidates by pitch-class, serial, rhythmic, gestural, and range diagnostics changes adherence while the trained checkpoint and test requests remain fixed.

The approach studied here separates probabilistic generation from deterministic score analysis. A causal Transformer samples several candidates, after which post-tonal functions rerank them by pitch-class coverage, interval-vector distance, row order, aggregate completion, rhythmic profile, gesture features, and instrumental range. This design accepts the computational cost of multiple candidates in exchange for diagnostics that can be shown to a composer and saved with the score.

2.4 Position of the present study

The study connects three established practices: formal symbolic manipulation from computer-assisted composition, event-based neural sequence generation, and constraint-aware selection. Its narrower contribution is an evaluated workflow conditioned on 4–16-measure post-tonal sketch requests that returns both MusicXML and an explanation of constraint adherence. The legal synthetic corpus makes the condition–target relation reproducible, while the controlled decoding comparison tests the contribution of reranking without changing the checkpoint or test material.

3 Methodology

3.1 Task formulation

The system generates a symbolic score fragment x from a user-specified condition bundle c. The bundle may contain a pitch-class set S, its six-dimensional interval vector $v(S)$, a twelve-tone row r, a serial form label $f \in \{P_n, R_n, I_n, RI_n\}$, a rhythmic-profile label, a gesture label, an instrument, a voice count, and a measure count. The target is a 4–16-measure event sequence that can be exported as MusicXML. No waveform or audio representation is used.

We represent the conditional generation problem as

$$p_\theta(x \mid c) = \prod_{t=1}^{T} p_\theta(x_t \mid c, x_{<t}), \tag{1}$$

where x_t is a score-event token and θ denotes the Transformer parameters. The practical objective is not only probable token continuation but also explicit satisfaction of score-level post-tonal constraints.

3.2 Legal synthetic corpus

Because much post-1945 art-music repertoire remains under copyright, the default corpus is generated procedurally rather than scraped from scores. Each sample draws from seven reference pitch-class sets, seven rhythmic profiles (sparse, medium, dense, additive, burst, sustained, and pointillistic), seven gesture labels, six instrument labels, 2–8 voices, and 4–16 measures. A twelve-tone row is included for all serial-focused samples and with probability 0.45 in the general corpus; its transformation label is sampled from the 48 $P/R/I/RI$ forms. Durations and onset shifts are quantized to a quarter of a quarter note.

The rule generator first constructs a rhythmic skeleton for each voice. Gesture-specific operations alter onset offsets, note omission, rest probability, duration, and register. Pitch classes are then drawn from the target set or cycled through the requested row form. Instrument-specific MIDI ranges constrain register selection. Onsets and durations use a sixteenth-note grid, represented as 0.25 quarterLength units. This process supplies reproducible condition–target pairs; it is not presented as a model of any individual composer or historical style.

3.3 Condition and score tokenization

Conditions are serialized as a prefix rather than encoded by separate neural subnetworks. The prefix begins with `BOS`, followed by zero or more `PC`, interval-vector, row-pitch, and row-form tokens, then rhythmic profile, gesture, voice-count, measure-count, and instrument tokens, and terminates with `SEP`. Explicit `NO_PCSET` and `NO_ROW` symbols represent absent pitch-set and serial conditions.

The score body uses `BAR`, `TIME_SHIFT`, `VOICE`, `PITCH`, `OCTAVE`, `DUR`, and `REST` events, followed by `EOS`. This compact representation preserves the information needed for quantized polyphonic MusicXML reconstruction while keeping sequences within a 256-token context. The training loss is standard teacher-forced negative log likelihood over the concatenated condition and score sequence:

$$\mathcal{L}_{\mathrm{NLL}}(\theta) = -\sum_{t=1}^{T} \log p_\theta(z_t \mid z_{<t}), \tag{2}$$

where $z = [c; x]$. Consequently, reported token accuracy is a full-sequence next-token diagnostic and should not be interpreted as a direct measure of musical quality.

3.4 Conditional Transformer

The neural generator is a causal Transformer [14] with six layers, hidden dimension 384, and six attention heads. Each layer uses a GELU feed-forward block of width $4d$, learned positional embeddings, pre-normalization, and dropout 0.1. A triangular attention mask prevents access to future tokens. The model contains no audio encoder, pretrained language model, or external score corpus.

At inference, top-k sampling with $k = 20$ produces one or more event-token continuations from the condition prefix. Tokens after `SEP` are parsed back into timestamped notes and rests. Instrument, gesture, and rhythmic metadata are retained for analysis and MusicXML export.

3.5 Constraint-guided candidate reranking

The proposed decoder draws K candidates and applies deterministic score-level analysis to each. Let $m_j(x, c)$ denote a penalty associated with constraint j. Candidate selection is

$$x^* = \arg\min_{x \in \mathcal{C}_K} \sum_j \lambda_j m_j(x, c), \tag{3}$$

where $\mathcal{C}_K$ is the sampled candidate set. The implemented weights are 1.0 for missing pc-set coverage, 0.05 for interval-vector distance, 1.0 for row-order error, 0.5 for incomplete aggregate coverage, 0.5 for rhythmic-profile distance, 0.5 for gesture inconsistency, and 2.0 for instrument-range violations. Pc-set, interval-vector, and row-order terms are inactive when their targets are absent. Aggregate completion remains active for every sample. This design choice can oppose a small target set by rewarding additional pitch classes, so its consequences are measured rather than assumed beneficial.

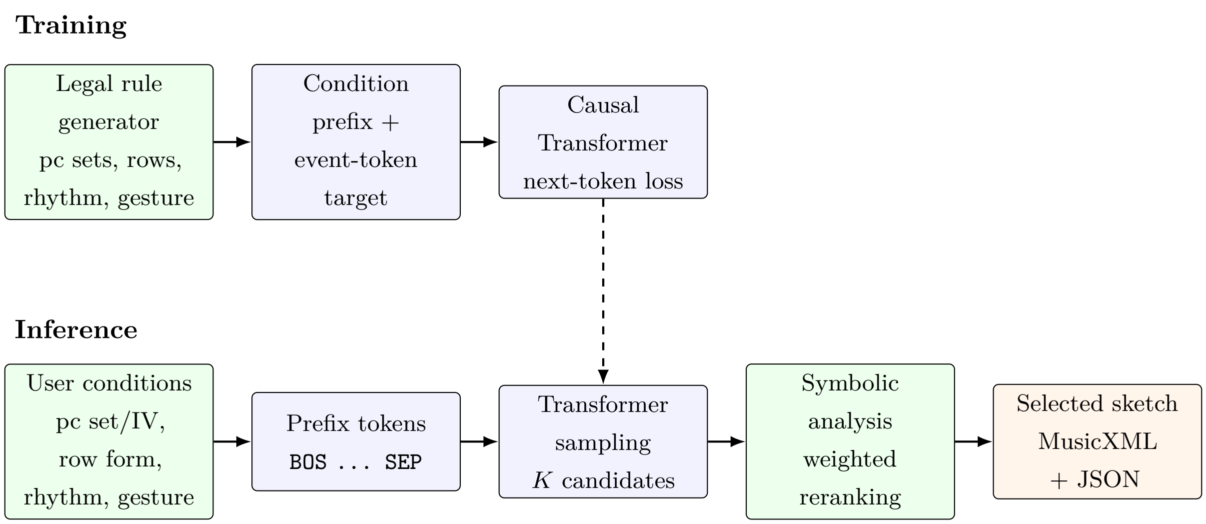

Figure 1: System overview. A legal procedural generator supplies condition–score pairs for causal language-model training. At inference, the same condition prefix produces K candidate event sequences. Non-differentiable post-tonal and notational diagnostics rerank candidates before MusicXML and an analysis report are exported.

All terms are used only for inference-time reranking; they are not presented as differentiable training losses.

The controlled experiment uses $K = 1$ as the single-candidate decoder and $K = 4$ as the constraint-reranked decoder. Both use the same trained checkpoint, test conditions, and per-sample random seed, so the first candidate in the $K = 4$ set is identical to the $K = 1$ output. Training data, model parameters, and test conditions are therefore held fixed; candidate count and symbolic selection define the planned contrast.

3.6 Explainable score analysis and export

Each selected candidate receives a JSON report containing pc-set coverage, generated interval vector, interval-vector distance when applicable, cyclic row-order accuracy, aggregate completion, per-measure density, rhythmic-profile distance, density-curve error, gesture features, gesture consistency, register spread, and instrument-range violations. Serial transformation accuracy currently uses the same cyclic order comparison as row-order accuracy and is therefore treated as an alias rather than an independent endpoint.

MusicXML is generated with score parts, measures, voices, pitches, rests, and durations [16]. Export success requires both file creation and structural parsing as a `score-partwise` or `score-timewise` document. Figure 1 summarizes the training and inference paths.

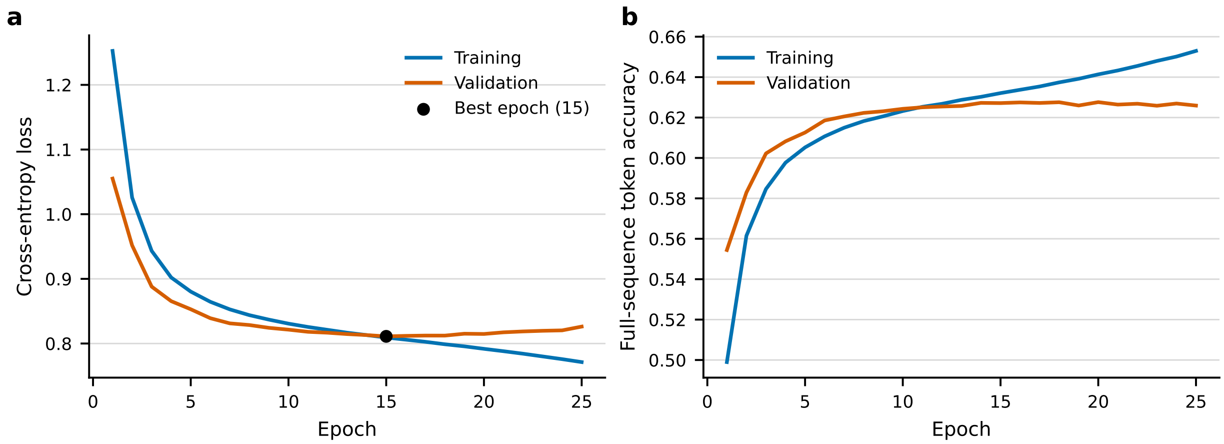

Figure 2: Primary-checkpoint training diagnostics from the saved run history. (a) Cross-entropy loss. The marker identifies the checkpoint selected at epoch 15. (b) Full-sequence next-token accuracy, which includes the condition prefix and is not a measure of musical quality.

4 Experimental Setup

4.1 Data splits

The general synthetic corpus contains 24,000 fragments: 20,000 for training, 2,000 for validation, and 2,000 for testing. Split membership is written explicitly into the processed data file rather than reconstructed at evaluation time. The primary corpus uses random seed 42 and conditions each target on a requested span of 4–16 measures, 2–8 voices, one of six instrument labels, one of seven rhythmic profiles, and one of seven gesture labels. The saved split summary records `smoke=false`; CPU smoke data are used only for software verification and never enter the reported experiment.

4.2 Training protocol

The Transformer has six layers, hidden size 384, six heads, feed-forward size 1,536, dropout 0.1, and a maximum sequence length of 256. Training uses AdamW with learning rate 3×10^{-4}, weight decay 0.01, batch size 16, gradient clipping at norm 1.0, and fp16 autocast. The maximum is 60 epochs with early-stopping patience 10. The primary checkpoint stopped after 25 epochs; its lowest validation loss, 0.8111, occurred at epoch 15. Dataloading uses zero worker processes for Windows compatibility. Figure 2 reports the saved optimization trajectories.

All full training ran locally with Python 3.11.9 and PyTorch 2.5.1+cu121 on an NVIDIA GeForce RTX 4060 Ti with 16GB VRAM. CUDA was available throughout the completed run. No CUDA out-of-memory adjustment or dataset reduction was used.

4.3 Independent-seed training replication

Three memory-safe replication runs used seeds 42, 43, and 44 on the same saved split of 20,000 training, 2,000 validation, and 2,000 test fragments. The processed corpus was held fixed, so the seeds changed model initialization and minibatch order rather than corpus membership. Each run used physical batch size 8 with two gradient-accumulation steps, preserving an effective batch size of 16. Architecture, optimizer, fp16 use, gradient clipping, and early-stopping patience matched the primary protocol. The best checkpoint from each run was evaluated teacher-forced on all 2,000 test fragments. We report the mean and sample standard deviation across the three runs. This replication measures variation in sequence-model optimization and does not repeat the $K = 1$ versus $K = 4$ generation comparison across training seeds.

4.4 Controlled decoding comparison

The primary comparison focuses on the inference procedure. Both conditions load the same seed-42 checkpoint and the same 2,000-item test split. The single-candidate condition samples $K = 1$ continuation. The proposed condition samples $K = 4$ continuations and selects the minimum-penalty candidate using Eq. (3). Sampling uses temperature 1.0, top-$k = 20$, and at most 224 new tokens. Evaluation seed 42042 is expanded deterministically as seed $42042 + i$ for test item i. Thus, the $K = 1$ candidate is also the first member of the corresponding $K = 4$ set. Per-item reports are saved for paired differences, win–tie–loss rates, and bootstrap confidence intervals.

4.5 Procedural reference and exploratory ablations

The rule generator is reported as a procedural reference because it constructs targets directly from the requested constraints. It is not an independent learned baseline and its serial accuracy should be understood as an oracle-style property of the generator.

The completed experiment archive also contains twelve trained neural configurations and one rule-reference row. These include independently trained vanilla and no-constraint models, explicit removal of pc-set or row information, focused pc-set/serial/rhythm/gesture configurations, and fixed-default variants. In the latter, rhythm or gesture variability is collapsed to `medium` or `fragmented`; those configurations do not remove the corresponding token type. Because configurations use different corpus seeds, their aggregate differences are treated as exploratory rather than as paired causal effects.

Two independently seeded no-constraint configurations are retained as replication-style diagnostic rows. Metrics whose target condition is absent are marked not applicable rather than interpreted as perfect performance.

4.6 Score and expert-evaluation outputs

For each evaluated system, 20 examples are exported for structural MusicXML checking. The proposed-system package contains paired MusicXML and JSON reports together with a blind rating form covering post-tonal coherence, pc-set/interval consistency, serial clarity, rhythmic control, gesture recognizability, notational usefulness, and usefulness for contemporary composition sketching. No human ratings had been collected at the time of this automatic evaluation; corresponding artistic claims remain pending.

5 Evaluation Metrics

Metrics are divided into sequence-model diagnostics, condition-adherence measures, and score-output checks. Arrows in the result tables indicate whether larger (↑) or smaller (↓) values are preferred.

Next-token diagnostics. Token accuracy is the fraction of non-padding targets whose next token is predicted by the maximum-logit class. Cross-entropy is averaged over the same concatenated condition–score sequence. Because condition-prefix positions are included, these values diagnose language-model fit but are not musical-quality scores.

Pitch-class-set measures. For generated pitch-class support $P(x)$ and nonempty target set S, coverage is

$$\mathrm{Cov}_{\mathrm{pc}}(x, S) = \frac{|P(x) \cap S|}{|S|}. \tag{4}$$

This is recall-like and does not penalize pitch classes outside S. Interval-vector distance is the L_1 distance between the generated and target six-entry interval-class vectors. The metric is not applicable when no target set or interval vector is supplied.

Serial measures. Cyclic row-order accuracy compares every generated note in global decoded event order with the corresponding position of the requested row form, wrapping after 12 positions. Simultaneous notes are ordered by their token sequence rather than treated as unordered verticalities. Aggregate completion is $|P(x)|/12$. The implementation's serial-transformation accuracy is numerically identical to cyclic row-order accuracy; to avoid double-counting, the manuscript reports only row-order accuracy. Row-conditioned metrics are not applicable when no row and form label are supplied.

Rhythmic measures. Each fragment is converted to a per-measure note-onset density curve. Density-curve error is the mean absolute difference from a deterministic reference realization generated with seed 1234. This reference is a profile template, not the exact stochastic rhythm realization used when constructing a corpus target. Rhythmic-profile distance normalizes that error by the maximum reference density, allowing comparisons across profile types and lengths. Lower values indicate closer agreement with the fixed template.

Gesture measures. Each gesture label is associated with expected ranges of within-measure onset dispersion, register spread, note density, mean duration, and rest-duration mass normalized by fragment span. The last quantity can exceed one in polyphonic passages because overlapping voices contribute separately. Feature-wise closeness values are averaged to produce a score in $[0, 1]$. This is an explainable heuristic consistency measure rather than a perceptual judgment.

Notation and range checks. Range-violation rate is the proportion of pitched events outside the configured instrument range. MusicXML export success is the proportion of requested exports that are written and parsed structurally as MusicXML scores. Structural success does not imply engraving quality; that question is reserved for expert evaluation.

For the controlled comparison, aggregate values are computed over all 2,000 test conditions and paired per-item records are retained for mean differences, win–tie–loss rates, and paired percentile-bootstrap confidence intervals. Endpoint intervals are reported without a multiple-comparison adjustment.

6 Results

6.1 Controlled same-checkpoint decoding

The controlled analysis paired all 2,000 test conditions, comprising 914 row-conditioned and 1,086 non-serial fragments. The fixed checkpoint had full-sequence token accuracy 0.6254 and cross-entropy 0.8158; these next-token diagnostics are identical across the decoding conditions because no retraining is involved. Table 1 reports per-condition differences between the first sampled candidate ($K = 1$) and four-candidate symbolic reranking ($K = 4$).

Reranking raised cyclic row-order accuracy on serial conditions from 0.1580 to 0.2350, a favorable mean difference of 0.0769 (paired bootstrap 95% CI [0.0705, 0.0833]). It selected a better row-order candidate for 65.4% of serial requests, tied for 25.4%, and selected a worse candidate for 9.2%. Across all conditions, rhythmic-profile distance decreased from 0.8712 to 0.8159 (favorable difference 0.0553; [0.0493, 0.0618]), density-curve error decreased by 0.2332 ([0.2082, 0.2584]), and gesture consistency increased from 0.5246 to 0.5535 (0.0289; [0.0249, 0.0330]). Instrument-range violations were already rare but decreased from 0.00055 to 0.00009.

Interval-vector distance decreased by 0.5075 overall ([0.3640, 0.6655]). The subgroup means expose the different scales of the task: non-serial distance decreased from 0.2716 to 0.0166, whereas

Metric	Subset	$K = 1$	Reranked $K = 4$	Favorable Δ	95% CI	n
PC coverage	all	0.9954	0.9952	-0.0003	[-0.0021, +0.0014]	2000
PC coverage	non-serial	0.9946	0.9993	+0.0047	[+0.0028, +0.0068]	1086
PC coverage	serial	0.9965	0.9903	-0.0062	[-0.0094, -0.0034]	914
IV distance	all	26.2430	25.7355	+0.5075	[+0.3640, +0.6655]	2000
IV distance	non-serial	0.2716	0.0166	+0.2551	[+0.1924, +0.3223]	1086
IV distance	serial	57.1018	56.2943	+0.8074	[+0.5153, +1.1379]	914
Row accuracy	serial	0.1580	0.2350	+0.0769	[+0.0705, +0.0833]	914
Aggregate	all	0.6634	0.6597	-0.0037	[-0.0052, -0.0022]	2000
Aggregate	serial	0.9963	0.9892	-0.0071	[-0.0101, -0.0044]	914
Aggregate (diagnostic)	non-serial	0.3832	0.3824	-0.0008†	[-0.0020, +0.0005]	1086
Rhythm distance	all	0.8712	0.8159	+0.0553	[+0.0493, +0.0618]	2000
Density error	all	5.8896	5.6563	+0.2332	[+0.2082, +0.2584]	2000
Gesture score	all	0.5246	0.5535	+0.0289	[+0.0249, +0.0330]	2000
Range violation	all	0.0006	0.0001	+0.0005	[+0.0003, +0.0007]	2000

Table 1: Controlled same-checkpoint decoding comparison. Favorable differences are oriented so that positive values favor four-candidate constraint reranking. Confidence intervals are paired percentile bootstrap intervals over test conditions and are not adjusted for multiple endpoints. †The non-serial aggregate row is a raw diagnostic change because no aggregate target is present for that subset.

serial distance decreased from 57.1018 to 56.2943 because row-conditioned fragments traverse the aggregate while retaining a smaller pc-set interval-vector target.

The pitch and aggregate results show a trade-off rather than uniform improvement. Overall pc-set coverage remained near ceiling and changed from 0.9954 to 0.9952; its interval included zero (favorable difference -0.0003; $[-0.0021, 0.0014]$). Coverage increased by 0.0047 in the non-serial subset but decreased by 0.0062 in the serial subset. Serial aggregate completion decreased from 0.9963 to 0.9892 (favorable difference -0.0071; $[-0.0101, -0.0044]$). The non-serial aggregate value is reported only as a diagnostic because those conditions have no twelve-tone aggregate target. No multiplicity correction was applied to the endpoint-wise bootstrap intervals; they are therefore descriptive uncertainty intervals rather than a family-wise inferential claim.

Figure 3 summarizes prespecified favorable relative changes. The near-ceiling pc-set endpoint and rare range violations produced high tie rates, while row, rhythm, and gesture selection altered a larger proportion of candidates. The large relative change in non-serial interval-vector distance reflects its small $K = 1$ baseline and should be read together with the absolute values in Table 1.

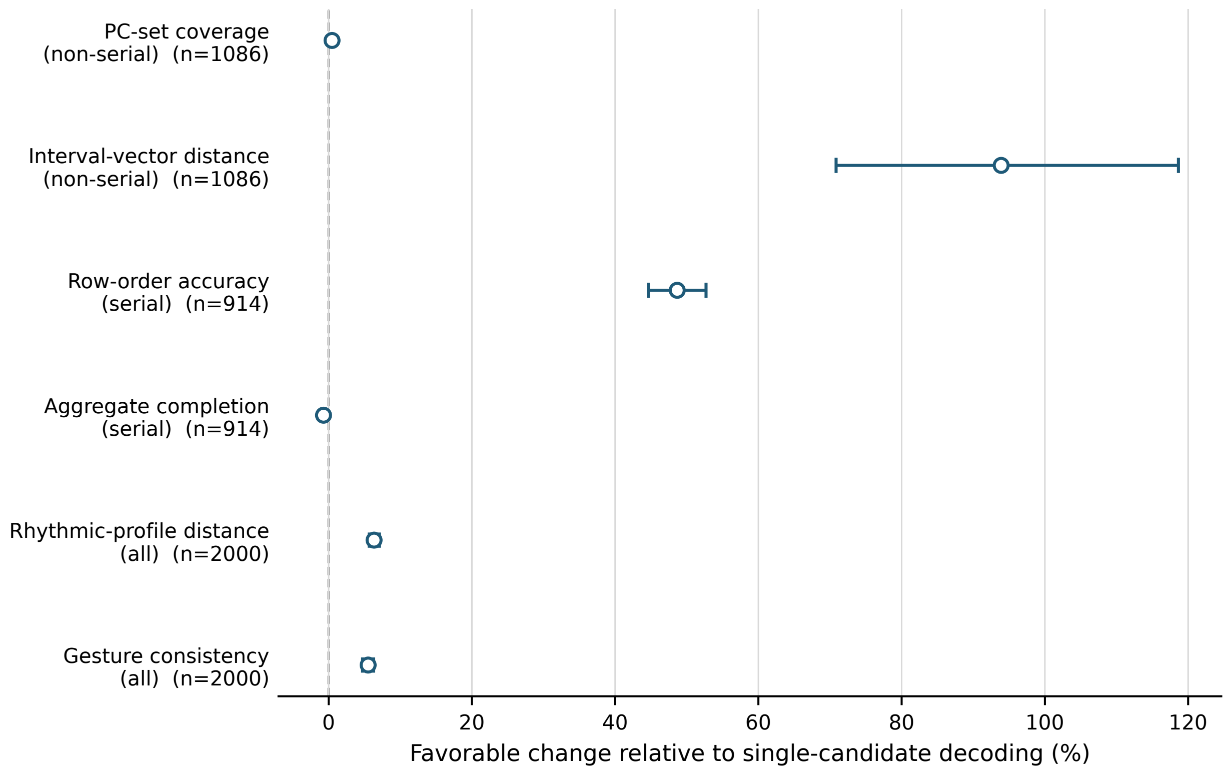

Figure 3: Controlled effects of four-candidate reranking relative to single-candidate decoding. Points are favorable mean changes as a percentage of the $K = 1$ mean; bars are paired bootstrap 95% intervals. Positive values favor reranking. Aggregate completion is negative because serial completion decreased. Intervals are not adjusted for multiple endpoints.

6.2 Independent-seed sequence diagnostics

The three memory-safe replications stopped after 25–26 epochs, with their selected checkpoints at epochs 15–16. Mean best validation loss was 0.8119±0.0011. On the common 2,000-fragment test split, teacher-forced token accuracy was 0.6259±0.0005 and cross-entropy was 0.8160±0.0012 (mean ± sample SD, $n = 3$; Table 2). These values show similar sequence-prediction diagnostics under the three initialization and shuffle seeds. They do not estimate cross-seed variation in constraint-guided generation because the full paired $K = 1$ versus $K = 4$ decoding experiment was run only with the primary checkpoint.

6.3 Exploratory configuration archive

Tables 3 and 4 retain the completed procedural reference and independently trained neural configurations. They document that all intended configurations ran on the full 20,000/2,000/2,000 corpus regime and produced 2,000-sample evaluation rows. These rows use different generated

Seed	Epochs	Best epoch	Val. loss	Test token acc.	Test model loss	Eff. batch
42	26	16	0.8111	0.6265	0.8149	16
43	25	15	0.8131	0.6255	0.8173	16
44	26	16	0.8114	0.6259	0.8157	16
Mean ± SD	–	–	0.8119 ± 0.0011	0.6259 ± 0.0005	0.8160 ± 0.0012	–

Table 2: Independent-seed training replication on a fixed synthetic corpus. Test metrics are teacher-forced; constraint-guided decoding is evaluated separately in the controlled seed-42 experiment.

Experiment	Token acc.	PC cov.	Row acc.	Rhythm dist.	Gesture	XML
Rule reference	–	0.9968	1.0000	1.7150	0.4025	1.0000
Independent vanilla	0.6279	0.9967	0.1579	0.9093	0.5278	1.0000
Original guided	0.6254	0.9949	0.2320	0.8163	0.5564	1.0000

Table 3: Archived aggregate results. Independent neural rows use different generated corpora and are descriptive rather than paired comparisons. Row accuracy is averaged only over row-conditioned samples, so its denominator differs from the 2,000-item test-set total and can vary by configuration.

corpora and seeds, however, so their aggregate differences are descriptive and are not used to estimate the causal effect of reranking. Missing cells denote an absent or fixed target, not perfect adherence.

6.4 MusicXML output and representative fragment

All 20 controlled reranked examples written for inspection parsed as `score-partwise` MusicXML. This is a structural check, not an assessment of engraving or compositional quality. Figure 4 shows controlled example 019, selected because every part contains the requested five measures. The request specifies five clarinet voices, pc-set $\{0, 3, 4, 7, 8, 11\}$, row form RI_6, a sustained rhythmic profile, and a fragmented gesture. Its automatic report gives pc-set coverage 1.0000, interval-vector distance 51.0000, row-order accuracy 0.2273, aggregate completion 1.0000, rhythmic-profile distance 1.6000, gesture consistency 0.3886, and range-violation rate 0.0000. These values illustrate the coexistence of satisfied and weak constraints in one editable output rather than a best-case musical claim.

7 Discussion

The controlled comparison answers a narrow question: with model parameters, test requests, and the first random candidate held fixed, symbolic reranking changes measured constraint adher-

Figure 4: Representative constraint-reranked MusicXML fragment (controlled example 019). Requested and written span: 5/5 measures; instrumentation: five clarinet parts; conditions: pc-set $\{0, 3, 4, 7, 8, 11\}$, RI_6, sustained rhythm, and fragmented gesture. The rendered notation is included for inspectability; no expert quality rating has yet been collected.

Experiment	Token acc.	PC cov.	Row acc.	Rhythm dist.	Gesture	XML
No constraints (seed 44)	0.6242	–	–	–	–	1.0000
No pc-set token	0.6176	–	0.1588	0.8726	0.5218	1.0000
No row token	0.6181	0.9931	–	0.8725	0.5283	1.0000
Fixed rhythm	0.6058	0.9982	0.1599	–	0.5364	1.0000
Fixed gesture	0.6556	0.9961	0.1552	0.8431	–	1.0000
Serial only	0.6593	–	0.4416	–	–	1.0000
PC-set only	0.6351	0.9982	–	–	–	1.0000
Rhythm only	0.6389	–	–	0.8824	–	1.0000
Gesture only	0.5822	–	–	–	0.5356	1.0000
No constraints (seed 53)	0.6239	–	–	–	–	1.0000

Table 4: Exploratory condition and fixed-default configurations. Dashes denote metrics that are not applicable because the corresponding target is absent or fixed. Row accuracy is averaged only over row-conditioned samples.

ence. The largest broadly distributed effect concerns serial order: 65.4% of serial requests improved, compared with 9.2% that worsened. Rhythm and gesture also changed often enough to shift their means and paired intervals in favorable directions. This supports the practical role of candidate selection as an inference-time control mechanism; it does not establish a new neural architecture or a better language model.

The results also reveal conflicts inside the implemented objective. Row order improved while serial aggregate completion and serial pc-set coverage decreased. General-corpus serial targets combine a twelve-tone row with a smaller pc-set and its interval vector, so complete row traversal, set restriction, and interval-vector matching cannot all be optimized without tension. Moreover, the selector minimizes a weighted sum rather than enforcing hard constraints. A candidate may therefore achieve a lower total penalty despite a worse component value. This behavior argues for exposing the full analysis report to the composer instead of reducing output quality to one scalar.

Near-ceiling pc-set coverage requires particular caution. The metric is recall-like, does not penalize off-set pitch classes, and tied in more than 97% of paired conditions. Its small subgroup changes should not be interpreted as pitch-set purity. The interval-vector improvement is more pronounced for non-serial material, but the large relative percentage arises from a low baseline. For serial material, the high absolute interval-vector distance reflects the mismatch between an aggregate-spanning output and a small-set target.

Reranking and evaluation deliberately reuse the same symbolic diagnostics. The paired changes therefore verify that the implemented selector can optimize its own inspectable objectives under a fixed candidate budget. They are not independent evidence of perceptual coherence, stylistic authenticity, or usefulness to a composer. Likewise, 20/20 structural MusicXML parses establish machine readability only. The representative score demonstrates an editable artifact and also shows why notation, measure-span adherence, ensemble balance, and compositional value require separate evaluation.

The three memory-safe training replications produced closely grouped teacher-forced test losses and token accuracies on the fixed corpus. Holding the corpus and effective batch size constant confines this comparison to initialization and minibatch-order variation. The result reduces concern that the reported sequence diagnostics arose from one failed optimization run, but it does not establish that the constraint-reranking effects persist across checkpoints. That claim requires repeating the paired generation protocol for each trained seed.

The independently trained configuration archive remains useful for software coverage and hypothesis generation, but differences between those rows cannot isolate individual constraints because corpus realization and random seed vary. The controlled decoder comparison is consequently the primary quantitative result. A stronger next study should repeat controlled decoding across the trained seeds, add a held-out constraint metric not used by the selector, measure pitch-class precision or Jaccard similarity and explicit measure- and voice-count adherence, and collect blind ratings from composers and post-tonal analysts. Until those data are collected, the present contribution is a reproducible, legally generated testbed and an explainable score-level control workflow rather than a claim of artistic superiority.

8 Limitations

The procedural corpus is both the principal strength and the principal boundary of this study. It makes the experiment legal, controlled, and reproducible, but its seven reference pc sets and one generation process do not represent the full distribution of contemporary compositional practice. The rule generator also supplies the training targets, so a learned model can recover its regularities without demonstrating transfer to independent human-authored music. Optional composer-provided MusicXML validation is therefore important future work.

Automatic metrics simplify richer musical concepts. Pc-set coverage is recall-like and does not penalize off-set pitch classes. Interval-vector distance is computed on the generated pitch-class support and can grow when material expands beyond a small target set. Rhythmic distance uses one fixed template realization rather than the exact stochastic rhythm used to construct each target. Gesture consistency uses hand-specified feature targets rather than perceptual labels, and its normalized rest-duration feature is not bounded by one in polyphony. Serial transformation accuracy currently aliases cyclic row-order accuracy. These measures are useful diagnostics, but no single value establishes artistic coherence.

The reranking objective applies aggregate completion to every sample. On non-serial samples, this term can reward pitch classes outside a small target set. The general corpus also combines a small pc-set condition with a twelve-tone row in serial samples; the rule target then traverses the row rather than remaining inside the smaller set. Full aggregate completion can therefore conflict with interval-vector matching in both the objective and serial target construction. The same diagnostics define the reranking score and the automatic endpoints, so a favorable controlled change measures successful optimization of those diagnostics rather than independent evidence of musical quality. No sensitivity analysis of the fixed penalty weights is available. We report serial and non-serial subsets separately, but the present design does not resolve this compositional trade-off.

The primary controlled evaluation holds one trained checkpoint fixed, while the broader configuration archive uses one training seed per configuration. Three proposed-model replications quantify variation in validation loss and teacher-forced test diagnostics, but the $K = 1$ versus $K = 4$ generation comparison was not repeated for those checkpoints. Cross-seed variation in constraint adherence therefore remains unavailable. Bootstrap intervals quantify paired test-condition variability only and are not corrected for examining multiple endpoints. The rhythm and gesture fixed-default experiments collapse label variability instead of removing those token types, and the two no-constraint rows are independent-seed replications rather than distinct architectures.

The event vocabulary favors compact quantized notation. It omits tuplets, complex meters, microtonality, extended techniques, articulation detail, beaming control, dynamics in the evaluated configuration, and engraving-level layout. One instrument label is repeated across all parts of a sample, so the current interface does not model mixed ensembles. Voice-count adherence is not scored, and the exporter writes only the requested part indices; generated events assigned to higher voice indices can therefore be omitted from MusicXML. Decoding is not grammar constrained, so incomplete or out-of-order event fields can be discarded by the parser. Sequence truncation at 256 tokens may also compress longer or denser requests. In the trained representation, a time-shift token is capped at 16 quarter-note beats and `BAR` tokens do not independently advance time during decoding, so longer silent spans can be shortened. Generation can also stop before the requested span or consume the token budget. Finally, the archived music21 export path creates measures only through the last decoded event and does not pad trailing empty measures. The number of serialized measures therefore does not consistently match the 4–16-measure condition. Structural MusicXML validity should not be equated with requested-length adherence, event preservation, or publication-quality notation.

Finally, the prepared blind expert evaluation has not yet been completed. The present results support claims about implemented symbolic constraints and file export, not claims about composer preference, perceptual coherence, or practical adoption. Those outcomes are not evaluated in the present study.

9 Reproducibility Statement

All bundled training data are generated from source code and fixed seeds; no copyrighted post-1945 score is downloaded or required. The full split and seed are recorded in `results/project2_full_split_summary.json`. Large processed tensors and checkpoints are excluded from version control, but the scripts and configuration files needed to regenerate them are retained.

On Windows, the CPU engineering check is executed with `scripts/smoke_project2.ps1`. The complete local CUDA pipeline is executed with `scripts/run_project2_full_local.ps1` in a Python 3.11 environment. Configuration files record architecture, optimizer, split sizes, fp16 use, early stopping, and the Windows-safe worker count.

Aggregate metrics are stored in `results/project2_metrics.csv` and `results/project2_constraints.csv`. Controlled metrics and per-sample reports use the `project2_controlled` prefix. Generated examples are paired with JSON analysis reports under `expert_eval/project2/`. Tables are created from CSV/JSON outputs rather than transcribed manually.

The independent-seed configs are stored under `configs/` and carry the suffixes `seed42`, `seed43`, and `seed44`. Per-seed rows and aggregate statistics are stored under `results/` as `project2_multiseed_training_metrics.csv` and `project2_multiseed_training_summary.json`. The table is regenerated by `post_tonal.analyze_multiseed_training`. This command verifies epoch histories, aligned 2,000-item test sizes, checkpoint hashes, and effective batch settings before writing outputs.

An anonymized source and data snapshot will be supplied during review in accordance with the journal's policy.

10 Conclusion

This study develops a reproducible neural-symbolic workflow for post-tonal composition assistance at the score level. Explicit pc-set, serial, rhythmic, gestural, register, and instrumentation conditions are serialized as prefix tokens for a locally trainable causal Transformer. Candidate continuations are inspected with post-tonal theory utilities and reranked before MusicXML and an explainable JSON report are exported.

The work is best understood as a controlled proof of concept rather than a model of contemporary style. Its technical value lies in making symbolic constraints measurable and editable while avoiding dependence on copyrighted repertoire. The controlled same-checkpoint experiment tests the contribution of inference-time reranking directly; the wider configuration archive documents behavior under removed or fixed-default conditions.

The next scientific steps are human evaluation by composers and analysts, cross-seed replication of the controlled $K = 1$ versus $K = 4$ decoding comparison, and validation on legally provided

MusicXML. Those additions are necessary before drawing conclusions about artistic usefulness, but the present system establishes the computational and evaluative substrate for that study.

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